JEFFREY MUTETHIA

SOFTWARE ENGINEER

Newcastle upon Tyne, UK

github.com/jeffreymutethia | linkedin.com/in/jeffreymutethia

EDUCATION

Newcastle University | MSc Advanced Artificial Intelligence | Expected Dec 2027

Whitman College | B.A. Computer Science | May 2023

EXPERIENCE

Software Engineer | Microsoft | Oct 2023 - Feb 2026

- Designed and operated backend services, ingestion pipelines and job orchestration for a large-scale security investigation graph, supporting 150% year-over-year data growth while improving validation, correctness and reliability.

- Built an automated query parity and diff framework across legacy and new distributed graph systems; migrated 20+ production queries and identified critical data-model gaps before cutover.

- Diagnosed and resolved a high-severity latency and routing regression in a Kubernetes production service; shipped structured logging, query telemetry and dashboards to improve debugging and incident recovery.

Software Engineer Intern | Microsoft | May 2023 - Aug 2023

- Built and deployed an OpenAI-integrated backend API and Microsoft Teams client adopted by 50+ security engineers, adding telemetry, monitoring and guardrails for a compliant production rollout.

Software Engineer Intern | Morgan Stanley | Jun 2022 - Aug 2022

- Built Java backend services for the Repo and Secured Financing trading desk, automating daily P&L workflows for 400+ trades and supporting event-driven workflow automation with Kafka.

- Expanded automated test coverage and added CI validation for production workflows, improving correctness and release reliability.

TECHNICAL SKILLS

Languages: Python, Java, C#, TypeScript, SQL

Engineering: Backend services, REST APIs, distributed systems, data structures and algorithms, debugging, automated testing, CI/CD

Infrastructure and data: Kubernetes, Docker, Azure, Linux, Kafka, Postgres, Redis, Neo4j, ingestion and job orchestration, observability

Applied AI: LLM-integrated applications, monitoring and guardrails